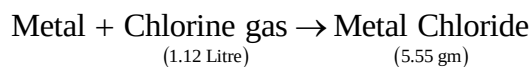


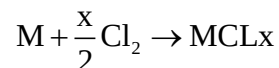
Solutions

Equivalent Concept

Question 1.



Let the metal chloride is MCl_x



Then n factor of chlorine



Then equivalent of chlorine Gas = equivalent of metal chloride.

$$\left(\frac{1.12}{22.4}\right) \times 2 = \frac{5.55}{\frac{M}{x} + 35.5}$$

$$\left(\frac{1.12}{22.4}\right) \times 2 = \frac{5.55}{\left(\frac{40}{x} + 35.5\right)}$$

$$\text{Then } \left(\frac{40}{x} + 35.5\right)(0.1) = 5.55$$

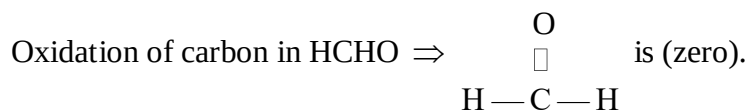
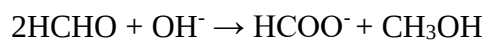
$$\frac{40}{x} + 35.5 = 55.55$$

$$\frac{40}{x} = 20.05$$

$$x = 2$$

Then valency of metal is 2.

Question 2.



Oxidation of carbon in $\text{HCOOH} = 2$

Oxidation of carbon in $\text{CH}_3\text{OH} = -2$

Then $\text{C}^0 \rightarrow \text{C}^{+2} + 2\text{e}^-$

$2\text{e}^- + \text{C}^0 \rightarrow \text{C}^{-2}$

Then total electrons used in this process is

$\Rightarrow$ for two moles 2 electrons are used, hence for one mole 1 electron is used hence n-factor is 1.

Hence mass of $\text{HCHO} = \frac{30}{1} = 30$

Hence $\frac{E}{10} = \frac{30}{10} = 3$

Question 3.

Solid element + Chlorine $\rightarrow$ Gas chloride
 $(\text{V}) \qquad \qquad \qquad \left(\frac{2\text{V}}{3}\right)$

Then milliequivalent of

Chlorine = milliequivalent of Gas chloride

Then $\left(\frac{\text{V}}{22.4}\right) \times 2 = \left[\frac{2\text{V}}{3 \times 22.4}\right] \times x$

$\Rightarrow$ n-factor for chlorine is 2

$\Rightarrow$ where (x) is the n-factor for gas chloride

$x = 3$

Then V.D. = 68.75

Then molecular mass = $68.75 \times 2 = 137.5$

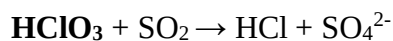
Then eqⁿ. mass of solid

$\frac{M}{x} + 35.5 = \frac{137.5}{3}$

Which is $= \frac{M}{x} = 10.33$

Then $\frac{12}{31} \times 10.33 = 4$

Question 4.



Then $\text{ClO}_3^- \rightarrow \text{Cl}^-$ [n-factor 6]

$\text{SO}_2 \rightarrow \text{SO}_4^{2-}$ [n-factor 2]

2 moles $\text{HClO}_3 \Rightarrow$ 6 moles SO_2

1 mole $\text{HClO}_3 \Rightarrow$ 3 mole SO_2

Then moles of $\text{SO}_2 = \left(\frac{V}{22.4} \right)$

$$\text{Moles of HClO}_4 = \frac{16.9}{84.5} = 0.2$$

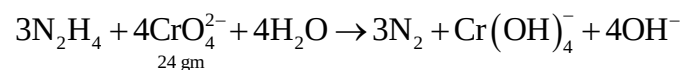
Then moles SO_2 used

$$0.2 \times 3 = \frac{V}{22.4}$$

$$V = 13.44$$

$$\text{Then moles in } 5V \Rightarrow \frac{13.44 \times 5}{22.4} \Rightarrow 3$$

Question 5.

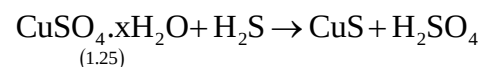


$$\text{moles of } \text{K}_2\text{CrO}_4 = \frac{24}{(78 + 52 + 64)} = 0.12$$

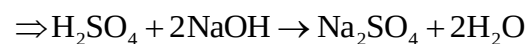
Then moles of $\text{N}_2\text{H}_4 = 0.0927$

Then mass of $\text{N}_2\text{H}_4 = 2.96$

Question 6.



$$\text{Then moles of } \text{H}_2\text{SO}_4 = \frac{1.25}{(159.5 + 18x)}$$



$$\Rightarrow \text{milimoles of } \text{H}_2\text{SO}_4 = \frac{10 \times 1}{2}$$

$$\Rightarrow \text{moles} = \frac{5}{1000}$$

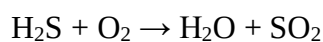
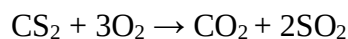
$$\text{Then } \Rightarrow \text{moles} = \frac{5}{1000}$$

$$= 1250 = 795.5 + 90x$$

$$90x = 454.5$$

$$X = 5.05$$

Question 7.

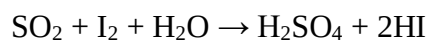


The moles of total gases which is collected

$$n = \frac{PV}{RT}$$

$$n = \frac{7.2 \times 82.1}{0.0821 \times 600}$$

$$n = 12$$



Then moles of I_2 = moles of SO_2 which is

$$\Rightarrow 5 \times 2 = 10$$

Then moles of CO_2 and $\text{H}_2\text{O} = 2$ moles

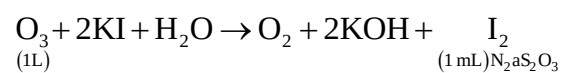
2 moles SO_2 gives 1 moles CS_2

Then 10 moles SO_2 gives $\Rightarrow 5$ moles CS_2

Rest gases (2 moles) also given by CS_2

Then total moles of = $5 + 2 = 7$

Question 9.



$$\frac{5.6}{5.6} = 1 \text{ NH}_2\text{O}_2 \text{ as it volume is 5.6.}$$

$$(NV)H_2O_2 = (NV)$$

$$\text{Then } (1 \times 12) = (NV)_{I^-}$$

$$(NV)_{I^-} = (NV)_{Na_2S_2O_3}$$

$$12 = 24 \times N$$

$$N = \frac{1}{2}$$

$$\text{And } I_2 = N \times V = \frac{9}{2} = (4.5)$$

$$\text{And moles of } I_2 = \frac{4.5}{2} \text{ [as It's n-factor is 2]}$$

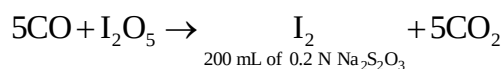
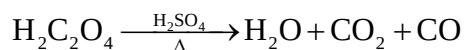
$$I_2 = 2.25$$

Question 10.

Moles of O_3 in the solution is same as mole of I_2

$$\text{Then It is } 2.25 \times 10^{-3}$$

$$\text{So } \% = \frac{2.25 \times 10^{-3}}{22.4} \times 100 = 5\%$$



$$\text{Then molecule of CO} = [200 \times 0.2]$$

$$\text{Then } I_2^{+10} \rightarrow I_2^0 \text{ (n - factor - 5)}$$

$$\text{Then m.moles} = \frac{200 \times 0.2}{5} = \frac{40 \times 0.2}{1000}$$

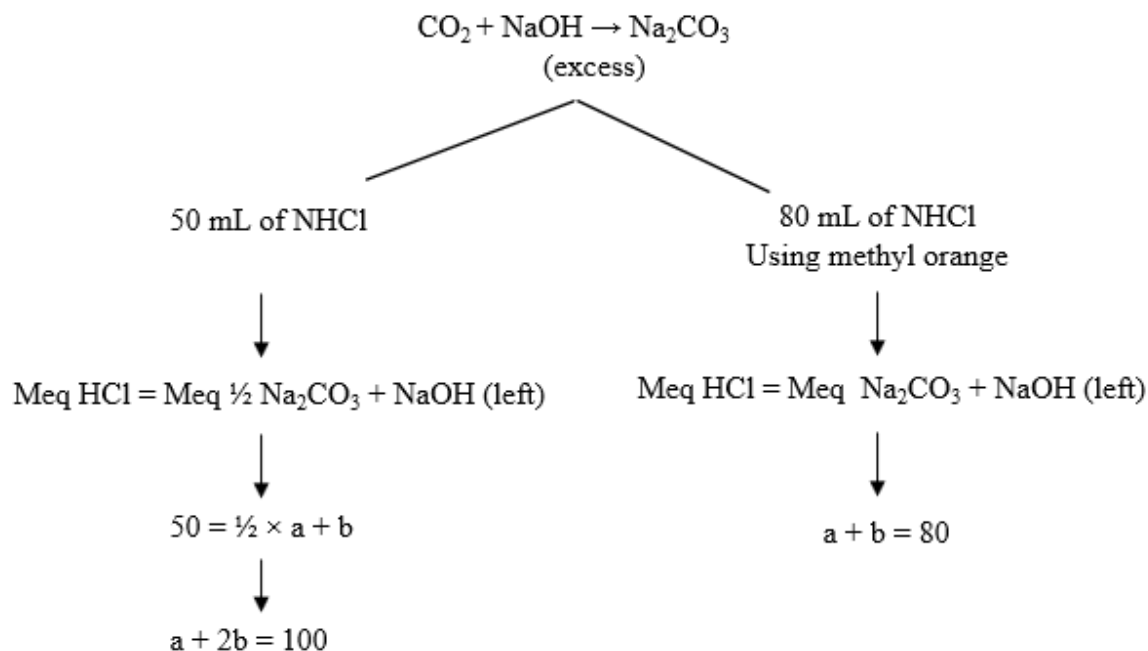
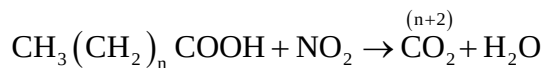
Then moles of oxalic acid

$$= \frac{40 \times 0.2}{1000}$$

$$\text{Oxalic acid mass} = \frac{40 \times 0.2}{1000} \times 90$$

$$= 0.729 \text{ gm}$$

Question 11.



Molecular mass of acid is $\text{CH}_3(\text{CH}_2)_n\text{COOH}$ is $(60 + 14n)$

On solving

$$\text{NaOH} = b = 20$$

$$\text{Na}_2\text{CO}_3 = a = 60$$

Then total meqⁿ Na_2CO_3 used = 60

The same eqⁿ is for CO_2 as well as for $\text{CH}_3(\text{CH}_2)_n\text{COOH}$

$$\text{Then } 60 = \frac{1.16}{(60 + 14n)} \times (n + 2)$$

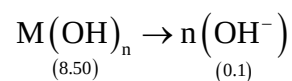
Then

$$60 \times (60 + 14n) = 1.16 \times (n + 2)$$

$$3600 + 840n = 1.16n + 2.32$$

$$n = 4$$

Question 12.



$$\text{moles of } M(OH)_n = \frac{8.50}{(M+17n)}$$

$$\text{then moles of } (OH^-) = \frac{n \times 8.50}{(M+17n)} = 1.7n$$

$$n \times 8.50 = 0.1 M + 1.7 n$$

$$6.8 n = 0.1 \times M$$

$$\frac{M}{x} = 68$$

Then equivalent mass of metal is 68.

Question 13.

$$Cu = x \text{ gm}$$

$$Ag = (2.7 - x) \text{ gm}$$

$\Rightarrow$ PP + was AgCl

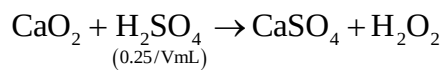
$$\% Cu = \frac{0.54}{2.7} \times 10 = 20\%$$



143.59 m contains $\rightarrow$ 1089 m Ag

$$2.87 \text{ gm} \rightarrow \frac{108}{143.5} \times 2.87 = 2.16 \text{ gm}$$

Question 14.



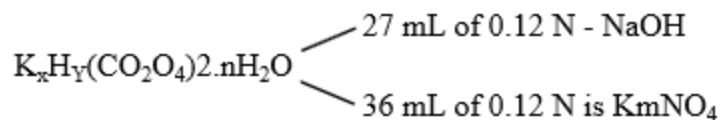
$$\text{Then moles of } CaO_2 = \frac{7.2}{72} = 0.1$$

Moles of CaO_2 = moles of H_2SO_4

$$0.1 = \frac{0.25 \times V}{1000}$$

$$V = 400 \text{ mL}$$

Question 15.



With NaOH

$$(\text{NV})_{\text{oxlate}} = (\text{NV})_{\text{NaOH}}$$

$$30 \times N = 27 \times 0.12$$

$$N = 0.108$$

And eqⁿ mass of oxlate

$$\frac{W}{E} = 0.108$$

$$\frac{9.15}{E} = 0.108$$

$$E = 84.72$$

Eqⁿ mass of oxlate with (KMnO₄)

$$30 \times 36 \times 0.12$$

$$N = 0.144$$

$$\frac{W}{E} = 0.144$$

$$E = 63.54$$

$$\text{And } x + y = 2Z$$

$$\text{Then } x : y : z$$

$$1 : 3 : 2$$

Then total equivalent mass should be equal and on solving

$$39 \times 1 + 1 \times 3 + 72 \times 2 + 18n = 294$$

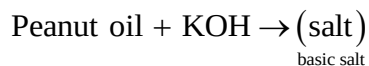
$$n = 2$$

Hence xyzn

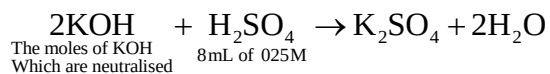
$$= 1 \times 3 \times 2 \times 2$$

$$= 12$$

Question 16.



This salt require 8 mL of 0.25 M H₂SO₄



Then KOH⁻ used = 10⁻²

= 8 m . moles

= 8 × 10⁻³ m. moles

= 448 × 10⁻³ gm KOH

= (448 mg KOH)

Then for 1 gm of peanut = $\frac{448}{1.5} = 298.66$

Question 17.

25 mL of under composed oxalic acid required

= 32mL 0.1 N KMnO₄ solution

Thus 250 mL of under composed oxalic acid requires

= 320 mL of 0.1 N KMnO₄ solution

= 32 mL of 1 N KMnO₄ solution

= 32 mL of 2 N oxalic acid solution

Oxalic acid used by solute

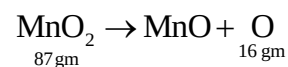
= (50 - 32) mL of 1N solution

= 18 mL of 1N solution

= 18 mL of MnO₂ solution

$$\text{mass of MnO}_2 = \frac{NEV}{1000} = \frac{1 \times 18 \times 87}{1000 \times 2} = 0.783$$

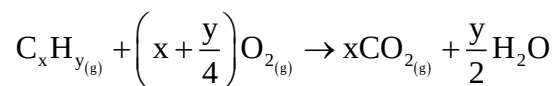
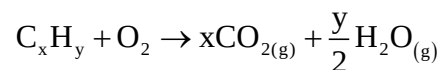
$$\text{Percentage of MnO}_2 = \frac{0.783}{1.6} \times 100 = 48.9$$



$$\text{Oxygen given by } 0.783 \text{ gm MnO}_2 = \frac{16}{87} \times 0.783 = 0.1449 \text{m}$$

$$\% \text{ if available oxygen} = \frac{0.144}{1.6} \times 100$$

Question 19.



$$\Rightarrow \text{vapour density} = \text{C}_x\text{H}_y = (12x + y) = 28$$

Molecular mass / 2

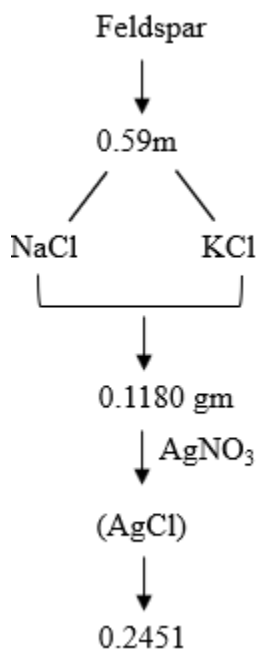
$$\Rightarrow 1 \text{ mole of } \underset{7.59\text{m}}{\text{C}_x\text{H}_y} \Rightarrow x \underset{159\text{m}}{\text{moles CO}_2}$$

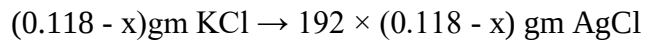
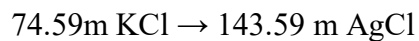
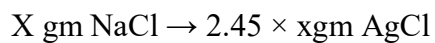
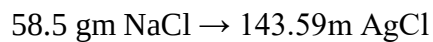
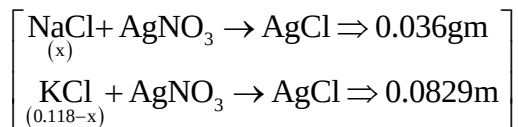
$$X = 2$$

$$Y = 4$$

Then formula of hydrocarbon is C₂H₄

Question 20.

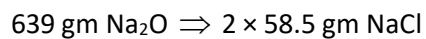
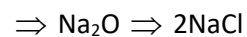




Then

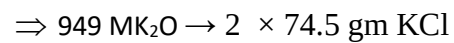
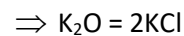
$$1.92 \times (0.118 - x) + 2.45 x = 0.2451$$

$$0.226 - 1.92x + 2.45x = x = 0.036$$



Then $\text{Na}_2\text{O} = 0.019 \text{ gm}$

$$\text{Then \% Na}_2\text{O} = \frac{0.019}{0.5} \times 100 = 3.8\%$$



$$\text{Then \% K}_2\text{O} = \frac{0.517}{0.5} \times 100 = 10.34\%$$